

Electricity & Circuits

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A **closed** (complete) circuit is an unbroken loop, so current flows and the bulb lights. An **open** circuit has a gap or an open switch, so no current flows. **Conductors** (metal) let current pass. **Insulators** (rubber, plastic, wood) stop it.

Round 1 — Retrieve It

1. What do we call a circuit that is a complete, unbroken loop?

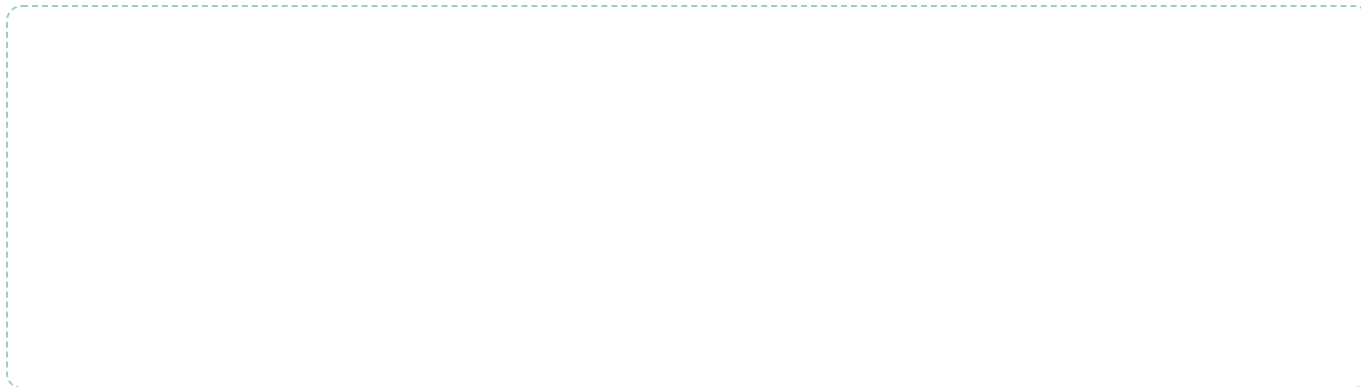
2. In an open circuit, does current flow? Will the bulb light?

3. What is a material called if it lets electric current pass through it?

4. Name one material that is an insulator.

Round 2 — Sketch It

Draw a simple circuit with a battery, a wire, and a bulb that is lit. Show the loop with no gaps. Label the battery, the wire, and the bulb.



Electricity & Circuits — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. A bulb in a circuit is not lighting up. You find that a wire has come loose from the battery. Is the circuit open or closed? Why?

2. You want to turn a light off without unplugging it. What part can you add to open and close the circuit?

3. Why are the wires in your home covered with rubber or plastic on the outside?

Round 4 — Reason from Evidence

A student builds a circuit and places a metal paper clip in the loop; the bulb lights. Then they swap the paper clip for a wooden craft stick and the bulb goes dark. What can the student conclude about metal and wood?

Electricity & Circuits — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain the difference between an open circuit and a closed circuit. Tell what happens to the bulb in each one.

C**Connect**

Connect the idea of conductors and insulators to a switch you use every day. Which part must be a conductor, and why?

E**Extend**

Plan a test to find out whether a new material is a conductor or an insulator using a battery, wires, and a bulb. Describe how you would know.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Electricity & Circuits

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. A closed circuit (a complete circuit).
2. No, current does not flow, so the bulb will not light.
3. A conductor.
4. Any: rubber, plastic, or wood.

Sketch It — look for

A complete loop with no gaps connecting the battery, wire, and bulb, all three labeled, and the bulb shown lit.

Apply It

1. Open — the loose wire makes a gap in the loop, so current cannot flow and the bulb stays dark.
2. A switch (it opens and closes the circuit).
3. Rubber and plastic are insulators; the covering stops the current from reaching people, which keeps them safe.

Reason from Evidence — look for

Metal is a conductor (it let current flow, so the bulb lit), and wood is an insulator (it stopped the current, so the bulb went dark).

ACE — Extend look-for

Plan places the new material in the loop of a working circuit; if the bulb lights, the material is a conductor, and if the bulb stays dark, it is an insulator.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.